

Control Theory - AeRoVe Documentation

Manan Garg

June 2025

1 Linear Control

1.1 Overview of Linear Systems

A linear system is a system with state $x \in \mathbb{R}^n$, that evolves in time according to the linear system of differential equations,

$$\dot{x} = Ax \quad (1)$$

Here $A \in \mathbb{R}^{n \times n}$ is called the **System Matrix** and $x \in \mathbb{R}^n$ is the **state vector**.

The solution to this linear system is given by -

$$x(t) = e^{At}x(0) \quad (2)$$

where $e^{At} \in \mathbb{R}^{n \times n}$ is defined as -

$$e^{At} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots \quad (3)$$

In some cases, we can **diagonalize** the matrix A , if it contains a set of eigenvectors that spans $\mathbb{R}^n$. Then by choosing the set of its eigenvectors as **basis** for the $\mathbb{R}^n$, the matrix representation of A in the new basis, which we call D , will be a diagonal matrix, whose diagonal entries are nothing but the eigenvalues of A .

Let $T \in \mathbb{R}^{n \times n}$ be the matrix whose columns are the eigenvectors of A . Then we have -

$$\begin{aligned} D &= \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) \\ A &= TDT^{-1} \end{aligned}$$

From this, we get -

$$e^{At} = Te^{Dt}T^{-1} = T\text{diag}(e^{\lambda_1 t}, e^{\lambda_2 t}, \dots, e^{\lambda_n t})T^{-1} \quad (4)$$

1.2 Stability of Linear System

The system is called **stable** if no component of x grows out of bound as time tends to infinity. On the other hand, if any component of x grows out of bound as time passes, the system is called **unstable**.

From equations (2) and (4), we can see that if all the exponential terms ($e^{\lambda_1 t}, e^{\lambda_2 t}, \dots, e^{\lambda_n t}$) are **decaying exponentials**, then the system will be stable. In other words, if all eigenvalues of the system have **negative real parts** ($\lambda_j = a_j + ib_j$, $a_j \leq 0$), then the system is stable. Moreover, as the real parts of the eigenvalues of the system become more negative, the system

will decay to zero faster.

Hence, we see that eigenvalues (also called poles) determine the nature of the dynamics of the system. Hence, it is of interest to us to be able to modify the eigenvalues of the system, and place them wherever we want on the complex plane. We need to place them in the negative-reals half-plane, in order to make the system stable.

When the system is stable, the state x will converge to 0. In this section we assume that the **target setpoint for x is 0**. It may be some non-zero $x^\dagger$ in the general case. Driving the state x to its target setpoint $x^\dagger$ is the objective of **Control Theory**.

1.3 Control Inputs to the System

We may have access to certain inputs to the system. By properly controlling these inputs, we can influence the dynamics of the state of the system. We call $u \in \mathbb{R}^q$, the **input vector**. In this case, the modified equations for the dynamics of the system are -

$$\dot{x} = Ax + Bu \quad (5)$$

where $u \in \mathbb{R}^q$, and $B \in \mathbb{R}^{n \times q}$.

It turns out that the nature of $u(t)$ that leads to optimal control of $x(t)$ is of the form -

$$u = -K_r x \quad (6)$$

where $K_r \in \mathbb{R}^{q \times n}$.

$$\dot{x} = Ax - BK_r x \quad (7)$$

$$\dot{x} = (A - BK_r)x \quad (8)$$

Hence, the modified system matrix becomes $(A - BK_r)$. Hence, by proper choice of K_r , we can change the eigenvalues of the system according to our will.

1.4 Controllability

The system (A, B) is called **controllable** if, with a suitable choice of the matrix K_r , we can place the eigenvalues of $(A - BK_r)$ *anywhere* on the complex plane.

If a system is controllable, we can reach *any* target state x , by appropriately choosing K_r .

The following tests are used to determine whether a system is controllable or not -

- **Controllability Matrix -**

Define the controllability matrix of the system (A, B) as -

$$\mathcal{C} = [B \ AB \ A^2B \ \dots \ A^{n-1}B] \quad (9)$$

The system (A, B) is controllable iff $\text{rank}(\mathcal{C}) = n$.

- **PBH Test -**

The system (A, B) is controllable iff $\text{rank}[(A - \lambda I)B] = n \ \forall \ \lambda \in \mathbb{C}$. Note, however, that we need only check this condition for λ that are eigenvalues of the system. For all other λ , this is always true.

Even if systems are controllable, some systems may be *more* controllable than other systems. Moreover, a system may be more controllable in some *specific direction* (in $\mathbb{R}^n$) than other directions.

If a system is more controllable, it means that for the same increase in input u , the state of the system x increases more. If it is more controllable in a specific direction, it means that the state x changes by a greater amount in that direction than other directions, for same increase in input u .

1.5 Linear Quadratic Regulator

While placing the poles of the system on the complex plane, there are two considerations we need to keep in mind -

- Placing poles with greater negative real part (leftwards on the complex plane) will cause faster convergence of system to the desired target.
- We can't place the poles too far left in the complex plane, because that would imply very high inputs u , which would cost us energy etc.

Keeping these factors in mind, we define a cost function -

$$J = \int_0^\infty (x^T Q x + u^T R u) dt \quad (10)$$

which is basically a time integral of a quadratic function of the state x and the input u . The first term $x^T Q x$ increases the cost if the state x is far from the desired setpoint. The second term $u^T R u$ increases cost if the input u is too high. The exact way in which J depends on the components of the x and u , is determined by matrices $Q \in \mathbb{R}^{n \times n}$ and $R \in \mathbb{R}^{q \times q}$.

For a given Q and R , there exists a single K_r that causes optimal control, minimizing the cost function corresponding to those Q and R . That K_r can be found using a technique called **Linear Quadratic Regulation**.

1.6 MATLAB Implementation

```
>> ctrb(A,B)
```

returns the controllability matrix for the system (A, B) .

```
>> rank(ctrb(A,B))
```

return the rank of the controllability matrix.

```
>> eig(A)
```

returns the eigenvalues of the matrix A .

```
>> K = place(A,B,eigs)
```

returns the K_r matrix so that eigs are the eigenvalues of the system matrix $(A - BK_r)$.

```
>> K = lqr(A,B,Q,R)
```

returns the optimal K_r calculated using linear quadratic regulation algorithm, for given values of Q and R .

2 State Estimation

2.1 Need for State Estimation

Till now, we wrote our system as –

$$\begin{aligned}\dot{x} &= Ax + Bu \\ u &= -K_r x\end{aligned}$$

Remember that u is the input to the system, that we are actively adjusting using the law $u = -K_r x$ in order to get the desired state x dynamics. We assumed in the above discussion that we **know**, i.e., can **measure** the state x .

However, that may not always be the case. In fact, most of the times, we are only able to measure a vector $y \in \mathbb{R}^p$, where

$$y = Cx \quad (11)$$

where $C \in \mathbb{R}^{p \times n}$. y is called the **measurement output vector**. The system **outputs** only y , and the state x is internal to the system. We can't **measure** x , but we want to **control** it.

Now, in case C is an invertible square matrix, then y effectively carries all the information that x carries, since we can get $x = C^{-1}y$.

However, it is also possible that $p < n$, which means that y does not carry the complete information of the **full state** x .

In that case, we need to implement a **state-estimation** algorithm that generates an estimate $\hat{x}$ of the state x .

The State-Estimator takes as input two sources of information to deduce the true state -

- Our model of the physical system, characterized by matrices A and B .
- The measurements y made on the system.

Hence, our system now becomes -

$$\dot{x} = Ax + Bu \quad (12)$$

$$y = Cx \quad (13)$$

$$u = -K_r \hat{x} \quad (14)$$

The estimate $\hat{x}$ is generated using the Kalman Filter algorithm, as is explained below.

2.2 Kalman Filter

The Kalman Filter is used to give the best estimate $\hat{x}$ to a state x , for a given system and measurement outputs. It is itself a linear dynamical system.

The Kalman Filter dynamics is given by -

$$\boxed{\frac{d}{dt}\hat{x} = A\hat{x} + Bu + K_f(y - C\hat{x})} \quad (15)$$

Later in section 2.4, we will give an intuitive interpretation of this equation. For now, let us rearrange the RHS in this equation, to get

$$\frac{d}{dt}\hat{x} = (A - K_f C)\hat{x} + Bu + K_f y$$

Now, let us define the error $\epsilon = x - \hat{x}$. Then we have -

$$\begin{aligned}\frac{d}{dt}\epsilon &= \frac{dx}{dt} - \frac{d\hat{x}}{dt} \\ &= Ax + Bu - (A - K_f C)\hat{x} - Bu - K_f Cx \\ &= (A - K_f C)(x - \hat{x}) = (A - K_f C)\epsilon\end{aligned}$$

Hence we have

$$\frac{d}{dt}\epsilon = (A - K_f C)\epsilon \quad (16)$$

Note the surprising result! This equation is completely analogous to the state-controller equation (8).

Hence, its mathematical analysis is exactly the same as before. ϵ decays to zero as $t \rightarrow \infty$, if this system is "stable", i.e., the eigenvalues of $(A - K_f C)$ all have negative real parts.

Moreover, we can choose K_f and hence change the eigenvalues of the system to whatever we want.

In the state controller, we are *driving* the state x to its target setpoint 0. In the state estimator, we are *driving* the error ϵ to 0, in other words, driving our estimate $\hat{x}$ to the true value x . Hence, as time progresses, our estimate $\hat{x}$ become closer and closer to the true value x .

2.3 Observability

The concept of Observability is, in a sense, dual to the concept of controllability.

A system (A, C) is called **observable**, if we can choose an appropriate K_f , so that the estimate $\hat{x}$ converges to the true value x as time tends to infinity. (In other words, $\epsilon \rightarrow 0$.)

Similar to controllability, the system is **observable**, if and only if, with an appropriate choice of K_f , we can place the eigenvalues of $(A - K_f C)$ *anywhere* we want on the complex plane.

Observability Matrix -

We can also define an **observability matrix**, analogous to the controllability matrix, that lets us test whether a given system is observable or not.

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \quad (17)$$

The system (A, C) is observable iff $\text{rank}(\mathcal{O}) = n$.

2.4 Interpreting the Kalman Filter Equation

As discussed previously, the state estimator uses two pieces of information - the **model**, and the **measurements**.

$$\frac{d}{dt}\hat{x} = \underbrace{A\hat{x} + Bu}_{\text{Model-Prediction}} + \underbrace{K_f(y - C\hat{x})}_{\text{Measurement-Update}}$$

In the Kalman Filter equation, the first term $A\hat{x} + Bu$ is the prediction according to the model, and our current estimate $\hat{x}$. It is also called the **a priori estimate**.

The second term $K_f(y - C\hat{x})$ is called the measurement update. It updates the a priori estimate using the new measurements made on the system. Basically, this term is proportional to the difference between the actual new measurement and the predicted measurement ($\hat{y} = C\hat{x}$). The final estimate is called the **a posteriori estimate**.

2.5 Linear Quadratic Estimator

The advantage of Kalman Filter, is that it gives us a way to find the *best* estimate of the state, even in the presence of noise, disturbances and model-uncertainties.

Noise can be classified into two kinds -

- Process Noise (w_d) - Refers to the noise/disturbances introduced in the actual system itself. It also includes inaccuracies in our model from the true system. E.g., a gust of wind pushes the drone slightly in some direction etc.
- Measurement Noise (v_d) - It refers to the noise in the measurement made by the sensor. E.g., Noisy measurement made by an IMU etc.

If we include the effect of noise, we can write our system as -

$$\dot{x} = Ax + Bu + w_d \quad (18)$$

$$y = Cx + w_n \quad (19)$$

We assume that both kinds of noise are Gaussian. Hence, we can determine the variances v_d and v_n of process and measurement noises respectively.

Given v_d and v_n , we might be more confident in our model than our measurements, or vice-versa. Hence, based on the relative confidence in model and measurements, we can choose an optimal K_f so that we give more "weight" to either the model-prediction term or the measurement-update term in the Kalman Filter dynamics.

Analogous to the state-controller, we can also define a cost function here, as

$$J = \mathbb{E}((x - \hat{x})^T(x - \hat{x})) \quad (20)$$

We can find out an optimal K_f that minimizes this cost function using an algorithm called **Linear Quadratic Estimator** (LQE).

2.6 MATLAB Implementation

```
>> obsv(A,C)
```

returns the observability matrix for the system (A, C) .

```
>> rank(obsv(A,C))
```

return the rank of the observability matrix.

```
>> K = lqe(A,B,v_d,v_n)
```

returns the optimal K_f calculated using linear quadratic estimation algorithm, for given values of v_d and v_n .